INJECTIVITY AND SURJECTIVITY OF QC MAPS CHECKED ON FLAT COVERINGS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Abelian structure on groupoid modules. Groupoids in Spaces, Lemma spaces-groupoids-lemma-abelian. Exactness detected on the presentation cover.
- Pullback and restriction of qc modules. Sheaves on Stacks, Lemma stacks-sheaves-lemma-quasi-coherent; Definition stacks-sheaves-definition-pullback. Compatibility of pullbacks with restrictions.
- Flat pullback reflects exactness. Standard scheme fact: flat pullback is exact; jointly surjective flat families reflect injective/surjective maps of modules (checked at stalks). Scheme-level reflection step.

Source reference. Original reference 06WU: sources/stacks-chow/tags.txt, source lines 8828–8828. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06VZ: sources/stacks-chow/tags.txt, source lines 8798–8798. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WK: sources/stacks-chow/tags.txt, source lines 8819–8819. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let I be a set and for $i \in I$ let $x_i : U_i \to \mathcal{X}$ be an object of $\mathcal{X}$. Assume that x_i is flat and $\coprod x_i : \coprod U_i \to \mathcal{X}$ is surjective. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be an arrow of $QCoh(\mathcal{O}_\mathcal{X})$. Denote φ_i the restriction of φ to $(U_i)_{étale}$. Then φ is injective, resp. surjective, resp. an isomorphism if and only if each φ_i is so.*

Proof. Choose a scheme U and a surjective smooth morphism $x : U \to \mathcal{X}$. We may and do think of x as an object of $\mathcal{X}$. This produces a presentation $\mathcal{X} = [U/R]$ for some groupoid in spaces (U, R, s, t, c) and correspondingly an equivalence

$$QCoh(\mathcal{O}_\mathcal{X}) = QCoh(U, R, s, t, c)$$

See discussion in Sheaves on Stacks, Section 06WU. The structure of abelian category on the right hand is such that φ is injective, resp. surjective, resp. an isomorphism if and only if the restriction $\varphi|_{U_{étale}}$ is so, see Groupoids in Spaces, Lemma 06VZ.

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For each i we choose an étale covering $\{W_{i,j} \to V \times_{\mathcal{X}} U_i\}_{j \in J_i}$ by schemes. Denote $g_{i,j} : W_{i,j} \to V$ and $h_{i,j} : W_{i,j} \to U_i$ the obvious arrows. Each of the morphisms of schemes $g_{i,j} : W_{i,j} \to U$ is flat and they are jointly surjective. Similarly, for each fixed i the morphisms of schemes $h_{i,j} : W_{i,j} \to U_i$ are flat and jointly surjective. By Sheaves on Stacks, Lemma 06WK the pullback by $(g_{i,j})_{small}$ of the restriction $\varphi|_{U_{étale}}$ is the restriction $\varphi|_{(W_{i,j})_{étale}}$ and the pullback by $(h_{i,j})_{small}$ of the restriction $\varphi|_{(U_i)_{étale}}$ is the restriction $\varphi|_{(W_{i,j})_{étale}}$. Pullback of quasi-coherent modules by a flat morphism of schemes is exact and pullback by a jointly surjective family of flat morphisms of schemes reflects injective, resp. surjective, resp. bijective maps of quasi-coherent modules (in fact this holds for all modules as we can check exactness at stalks). Thus we see

$$\varphi|_{U_{étale}} \text{ injective} \Leftrightarrow \varphi|_{(W_{i,j})_{étale}} \text{ injective for all } i, j \Leftrightarrow \varphi|_{(U_i)_{étale}} \text{ injective for all } i$$

This finishes the proof. □